SQL Query Analysis

1. Overall Workflow Performance Metrics

total_tasks	delayed_tasks	delay_rate	avg_completion_time	avg_cost_per_task
2500	2373	94.92	180.86	277.98

Insight:

- Total tasks processed: 2500
- 2373 tasks were delayed, giving a 94.92% delay rate.
- Average completion time is ~181 minutes.
- Average cost per task is ~278.

Key Insight:

The workflow system has **very high delays**, meaning most tasks are not completed within the expected time.

2. Processes With Highest Delays

process_name	total_tasks	delayed_tasks	delay_rate
Invoice Approval	480	462	96.25
Purchase Order	517	493	95.36
Customer Complaint	501	475	94.81
HR Onboarding	510	480	94.12
IT Support Ticket	492	463	94.11

Insight:

- Invoice Approval has the highest delay rate (96.25%).

- Followed by Purchase Order (95.36%) and Customer Complaint (94.81%).

Key Insight:

Finance-related processes like Invoice Approval and Purchase Orders are the biggest delay contributors.

3. Departments Causing Operational Inefficiencies

department	total_tasks	delayed_tasks	delay_rate
IT	497	474	95.37
Finance	504	480	95.24
HR	502	478	95.22
Operations	505	479	94.85
Customer Service	492	462	93.90

Insight:

- IT (95.37%) has the highest delay rate.
- Followed closely by Finance and HR (~95%).

Key Insight:

Operational inefficiency is distributed across departments, but IT and Finance contribute the most delays.

4. Employee Workload vs Delay Rate

	employee_workload	tasks	delay_rate
▶	1	252	96.43
	2	230	95.22
	3	240	92.92
	4	250	96.80
	5	271	94.83
	6	258	94.19
	7	255	94.90
	8	252	95.24
	9	255	93.33
	10	237	95.36

Insight:

- Employees handling higher workloads tend to have higher delay rates.
- Example: some employees handling 250+ tasks show ~96% delay rate.

Key Insight:

There may be workload imbalance, causing delays due to employee overload.

5. Tasks With Largest Time Overrun

	task_id	process_name	task_type	time_overrun
▶	TASK_2085	IT Support Ticket	Escalation	120
	TASK_1975	HR Onboarding	Data Entry	120
	TASK_559	Invoice Approval	Validation	120
	TASK_674	Customer Complaint	Approval	120
	TASK_1867	Purchase Order	Review	120
	TASK_1959	Invoice Approval	Review	120
	TASK_1264	Invoice Approval	Approval	120
	TASK_1128	IT Support Ticket	Escalation	120
	TASK_1134	HR Onboarding	Approval	120
	TASK_2060	Purchase Order	Escalation	120

Insight:

Major overruns occur in:

- IT Support Ticket – Escalation
- HR Onboarding – Data Entry
- Invoice Approval – Validation
- Customer Complaint – Approval

Key Insight:

Tasks involving approvals and escalations take significantly longer than expected.

6. Task Type Performance Analysis

task_type	total_tasks	avg_completion_time	delay_rate
Validation	493	185.42	96.15
Review	530	180.60	94.91
Data Entry	480	180.40	95.21
Escalation	504	179.43	94.25
Approval	493	178.51	94.12

Insight:

- Validation tasks have the highest delay rate (96.15%).
- Approval tasks have the lowest delay rate (~94%).

Key Insight:

Validation processes are the slowest step in the workflow pipeline.

7. Approval Level Bottlenecks

approval_level	total_tasks	avg_completion_time	delay_rate
Level 1	826	180.78	96.00
Level 2	821	180.45	94.64
Level 3	853	181.35	94.14

Insight:

- Level 1 approvals have the highest delay rate (96%).

- Levels 2 and 3 are slightly faster.

Key Insight:

The first approval stage is the main bottleneck, slowing down the entire process chain.

8. Cost Impact of Delays

delay_flag	avg_cost
1	277.93
0	278.88

Insight:

- Average cost for delayed tasks: ~277.9
- Average cost for non-delayed tasks: ~278.8

Key Insight:

Delays do not significantly change cost per task, meaning inefficiency is more time-related than cost-related.

9. Department With Highest Operational Cost

department	total_cost
Operations	141783.76
IT	140654.5
HR	139109.55
Finance	138633.01
Customer Service	134760.95

Insight:

- Operations department has the highest operational cost (~141,783).
- Followed by IT and HR.

Key Insight:

Operations consumes the largest share of operational resources.

10. Daily Task Volume

task_date	tasks_started
2024-01-01	9
2024-01-02	8
2024-01-03	8
2024-01-04	7
2024-01-05	11
2024-01-06	4
2024-01-07	2
2024-01-08	5
2024-01-09	7
2024-01-10	9
2024-01-11	9
2024-01-12	4
2024-01-13	8
2024-01-14	12
2024-01-15	7

Insight:

- Tasks are processed daily with relatively consistent volume.
- No major spikes or drops in the observed period.

Key Insight:

The workflow system has stable task inflow, so delays are not caused by sudden demand spikes.

11. Top 3 Employees With Highest Delay Rate

	assigned_employee_id	total_tasks	delayed_tasks	delay_rate	rank_delay
▶	EMP_35	20	20	100.00	1
	EMP_55	11	11	100.00	1
	EMP_140	16	16	100.00	1
	EMP_22	11	11	100.00	1
	EMP_51	13	13	100.00	1
	EMP_30	15	15	100.00	1
	EMP_86	16	16	100.00	1
	EMP_7	13	13	100.00	1
	EMP_16	23	23	100.00	1
	EMP_53	16	16	100.00	1

Insight:

Several employees show 100% delay rate, including:

- EMP_35
- EMP_55
- EMP_140
- EMP_16

Key Insight:

Certain employees consistently delay tasks, indicating performance or workload issues.

12. Department Contribution to Total Operational Cost

	department	department_cost	cost_percentage
▶	Operations	141783.76	20.4
	IT	140654.5	20.24
	HR	139109.55	20.02
	Finance	138633.01	19.95
	Customer Service	134760.95	19.39

Insight:

Cost distribution is fairly balanced:

- Operations: 20.4%

- IT: 20.24%
- HR: 20.02%
- Finance: 19.95%
- Customer Service: 19.39%

Key Insight:

Operational cost is evenly distributed across departments, with Operations slightly leading.

13. Detect Tasks With Significant Time Overrun

task_id	process_name	task_type	estimated_time_minutes	actual_time_minutes	overrun_minutes
TASK_559	Invoice Approval	Validation	90	210	120
TASK_674	Customer Complaint	Approval	88	208	120
TASK_917	Invoice Approval	Data Entry	49	169	120
TASK_1082	HR Onboarding	Escalation	24	144	120
TASK_1128	IT Support Ticket	Escalation	199	319	120
TASK_1134	HR Onboarding	Approval	207	327	120
TASK_1264	Invoice Approval	Approval	42	162	120
TASK_1299	HR Onboarding	Escalation	195	315	120
TASK_1436	HR Onboarding	Approval	162	282	120
TASK_1803	HR Onboarding	Review	86	206	120

Insight:

Several tasks exceed estimated time by ~120 minutes, such as:

- Invoice Approval – Validation
- Customer Complaint – Approval
- HR Onboarding – Escalation
- IT Support Ticket – Escalation

Key Insight:

The biggest time overruns occur in approval, escalation, and validation tasks, indicating process inefficiencies.